Mahabub Mollah

Principal Software Engineer

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SUMMARY

I am a Principal Software Engineer with 8+ years of experience in designing and implementing cloud-native applications. Skilled in various programming languages and frameworks, I have a proven track record of leading projects that enhance system performance and streamline workflows. My passion for mentoring engineers and delivering impactful solutions drives my commitment to excellence in software development.

SKILLS

Languages: JavaScript, TypeScript, PHP, Python, Go, SQL

Libraries & Frameworks: NodeJs, ExpressJs, NestJS, Laravel, CakePHP, Azure Function App (C#, .NET10), WordPress, Django, ReactJs, NextJs

Technologies & Tools: AWS (EC2, S3, Lambda, Dynamo db, App Sync), Azure(Blob, SignalR, Function App, Azure SQL, App Insight, Key Vault), Microsoft Power Platform, Docker, Kubernetes, Microservices, API Gateway, Elasticsearch, Microsoft Dataverse, PostgreSQL, MySQL, MongoDB, Kafka, RabbitMQ, GitHub Actions, CI/CD, IAM Governance, Secrets Management

Testing & QA: Jest, Mocha, Chai, PHP Unit, Cypress, Postman
AI Tools: GPT, Claude, Cursor AI, AWS Amplify AI

TECHNICAL ACHIEVEMENT

- Optimized Fabric → Dataverse sync with Azure Durable Functions and bulk Dataverse writes, cutting large-mill sync time by ~40–60% and repeat sync volume by ~50–90% via delta sync.
- Reduced average response time by 40% through query optimization, caching strategies, cursor-based pagination and efficient data pipelines in PostgreSQL/MySQL.
- Architected microservices and APIs with API Gateway and Saga patterns, enabling independent scaling and cutting deployment time by 50%
- Implemented Kafka pipelines with a push–pull model, reducing latency by 60% and scaling throughput to millions of events per second.
- Established CI/CD pipelines with Docker and Kubernetes, reducing manual deployment effort by 70%.
- Applied IAM governance, secrets management, and encryption standards, reducing security vulnerabilities by 35%.
- Built React/TypeScript UI with TanStack Query, lazy BOM loading, and Azure SignalR locks, reducing API calls by ~60–70% and improving large-table render speed by ~80–90%.
- Integrated AI-driven tools (GPT, Claude, Cursor AI) into engineering workflows, boosting developer productivity by 25%.

EXPERIENCE

Principal Software Engineer

BJIT(Bangladesh Japan Information Technology Limited)

05/2020 - Present, Dhaka

- Led cross-functional development teams through sprint planning, task prioritization, code reviews, and on-time delivery of backend and frontend features.
- Managed project execution end-to-end — requirement breakdown, risk handling, stakeholder coordination, CI/CD releases, and mentoring developers to improve team productivity and code quality.
- Designed and scaled microservices/APIs with API Gateway, Circuit Breaker, and Saga patterns.
- Implemented CI/CD pipelines, Docker/Kubernetes orchestration, and AWS cloud infrastructure
- Applied security best practices (IAM governance, secrets management, encryption standards

• Integrated AI-driven engineering tools (GPT, Claude, Cursor AI) to improve productivity

• Led backend/frontend development with Node.js, React, Next.js, PHP/Laravel

• Mentored engineers, conducted code reviews, and guided modernization efforts (monolith decomposition, API redesign, cloud migration)

• Technologies: AWS (EC2, S3, RDS, AppSync), Microsoft Azure, Kafka, RabbitMQ, MySQL, PostgreSQL, Express.js, NestJS, Laravel, CakePHP, Docker,Kubernetes.

Software Engineer

AntoPolis

01/2018 - March 2020, Dhaka

• Developed and maintained full-stack features using Node.js, React, Next.js, PHP/Laravel, applying design patterns (Factory, Observer, Strategy) to deliver clean, modular, and testable code.

• Optimized PostgreSQL/MySQL queries and schemas for performance and scalability, while ensuring reliability through code reviews, unit/integration testing, CI/CD, and production issue resolution on AWS (EC2, S3).

Junior Software Engineer

Future Track IT

11/2016 - 12/2017, Dhaka

• Designed and built scalable web applications using PHP, Laravel, MySQL, JavaScript, jQuery, and CSS, delivering maintainable features through structured development, testing, and debugging.

• Collaborated with cross-functional teams to align technical execution with business goals and ensure reliable, production-ready delivery.

PROJECTS

E-parts – Catalog management

04/2026 - Present, Dhaka

• Led full-stack delivery of a scalable mill data platform covering sync, BOM management, and real-time editing

• Built React/TypeScript UI with TanStack Query, lazy loading, and SignalR locks — ~60–70% fewer API calls, ~80–90% faster large-table render

• Built Azure Functions (.NET 10) with Durable Functions for Fabric → Dataverse sync — ~40–60% faster large-mill sync, ~50–90% less repeat sync data via delta sync

• Optimized Dataverse bulk writes with streaming pipelines; delivered BOM, Translations, Mill Search, Excel import, Activity Log, and Action Server integrations

• Tech Stack: React, TypeScript, MUI, TanStack Query, SignalR, .NET 10, Azure Functions, Fabric SQL, Dataverse, Azure Blob, Azure DevOps

Shin Sagantosu Fan Token – Store Discovery Platform

04/2025 -02/026, Dhaka

• Led full-stack development from design to deployment, delivering a scalable platform for 500+ stores

• Architected React/TypeScript frontend with Redux/React Query, ensuring state persistence and shareable filters

• Implemented Google Maps integration with custom clustering and resolved race condition bugs via ref-based tracking

• Built admin panel with CRUD, bulk CSV import, multi-image upload (AWS S3), and concurrent handling

• Implemented Google Maps integration with custom marker clustering, dynamic bounds fitting, and resolved race condition bugs using ref-based state tracking

• Designed API layer with debounced search, auto-complete, campaign filtering, caching, reducing API calls by 60%

• Integrated backend services with AWS Amplify,Dynamo db, RDS for high-availability workflows

• Owned CI/CD deployment workflows, optimizing cloud resources and reliability

• Tech Stack: React, NestJS, TypeScript, Redux, React Query, Tailwind CSS, Google Maps API, AWS (S3, Amplify,Dynamo db, RDS)

Mitsubishi Logistic – Warehouse Monitoring Application

11/2023 - 02/2025, Dhaka

- Built microservices backend (NestJS) processing 10M+ records/day, cutting batch time from 3h → 40m
- Integrated Kafka/RabbitMQ for async analytics, enabling near real-time productivity insights
- Applied API Gateway for unified access/authentication, reducing integration overhead by 30%
- Optimized performance service with caching/pipelines, lowering latency by 45% under peak loads
- Developed React/Redux dashboards for KPIs (throughput, downtime, utilization), reducing manual reporting by 70%
- Developed React-based monitoring dashboards and integrated Redux for predictable state management.
- Containerized with Docker, enabling independent scaling and fault-tolerant deployments
- Node.js, Express, React, Redux, Docker, API Gateway, Kafka, RabbitMQ, Microservices

A-B Will Vote – Online Voting & Social Platform

02/2022 - 04/2023, Dhaka

- Built Node.js/Express backend APIs to power voting, content management, and social features
- Developed React frontend components with Hooks/Redux for scalable state management.
- Delivered core features: multi-option questions, file/video uploads, social login (Facebook/LinkedIn), profile management, follow/unfollow, comments/replies, category-based search
- Enhanced admin panel with question sorting, category creation, pro-user management, and moderation tools
- Engineered buffered video streaming with back-pressure handling, ensuring smooth playback of large video files and preventing memory overload during peak usage
- Wrote unit/integration tests (Jest) ensuring reliability and maintainability
- Deployed a scalable platform combining online voting with social media functionality
- Technologies: Node.js, Express, React, React Hooks, Redux, AWS (EC2, S3), MySQL

BMS (Biomedical Management System) - Laboratory Management Platform

08/2020 - 02/2022, Dhaka

- Developed a - CakePHP 5.x system to manage material preservation, manufacturing workflows, and multi-facility operation
- Built modular backend APIs and components for approval workflows, material tracking, room reservations, and process step management.
- Designed interactive frontend with jQuery, Bootstrap, Data Full calendar and D3.js for scheduling, drag-and-drop reservations,and workflow analytics
- Implemented RBAC with granular permissions, QR/barcode tracking, PDF reporting, and multi-language support (English/Japanese)
- Migrated from Cake PHP 4.x → 5.x, ensuring PHP 8.1+ compatibility and improved performance
- Wrote PHP Unit test suites for controllers, models, and components, ensuring reliability and maintainability
- Technologies: CakePHP 5.x, PHP 8.1+, MySQL, jQuery, Bootstrap, D3.js, DataTables, FullCalendar, PHPUnit, RESTful APIs, i18n/l10n

EDUCATION

B.S.E in computer Science and Engineering

Dhaka International University (DIU)

08/2011 - 12/2017, Dhaka

CGPA: 3.60/4.00

CERTIFICATIONS

Node.js Developer Course
The Complete Node.js Developer Course (3rd Edition)

Web Application Development
Web Application Development course offered by BASIS

AWARDS

Best Performer of the Year
Recognized as Best Performer of the Year for outstanding contributions, technical excellence, and impactful delivery

Mentorship Role
Appointed as a mentor for web development, guiding a group of 30 students and professionals